

Automate Order Report Generation Using S3 & Lambda

Scenario

You work at a small e-commerce company. Every day, the operations team uploads a daily orders CSV file to S3. Currently, the manager manually opens the file, calculates total revenue per city, and emails the numbers to the sales team.

Your job is to **automate this** — the moment a CSV file is uploaded to S3, a Lambda function should automatically pick it up, generate a city-wise revenue summary, and save the report back to S3.

Data

A sample data file `orders.csv` is provided with this assignment. It contains 30 orders with the following columns:

```
order_id, customer_name, product, category, quantity, price, city,
order_date
```

Revenue for each order = `quantity × price`

What You Need to Do

Step 1 — S3 Setup

- Create an S3 bucket
- Create two folders inside it: `incoming/` and `reports/`
- `incoming/` is where the raw CSV gets uploaded
- `reports/` is where Lambda will write the output

Step 2 — Lambda Function

Create a Lambda function that does the following:

1. Gets triggered automatically when a `.csv` file is uploaded to the `incoming/` folder
2. Reads the CSV file from S3

3. Calculates **total revenue per city** (revenue = quantity × price)
4. Creates a new summary CSV with two columns: `city` and `total_revenue`, sorted by highest revenue first
5. Saves this summary file to the `reports/` folder with the name format:
`city_revenue_summary_<date>.csv`

Example output (city_revenue_summary_2026-03-10.csv):

```
city,total_revenue
Mumbai,7742.00
Bangalore,6553.00
Delhi,4699.00
...
```

Step 3 — Use a Lambda Layer

Since you'll use `pandas` to process the CSV, package it as a **Lambda Layer** and attach it to your function. Don't bundle `pandas` inside your function code.

Step 4 — Verify

- Upload the provided `orders.csv` to `incoming/`
- Check that the summary file appears in `reports/`
- Check CloudWatch logs to confirm the function ran successfully

Deliverables

Submit your work as a **zip file with documentation** or a **GitHub repository** containing:

- Your Lambda function code
- The sample data file
- A `README.md` explaining how to set everything up
- Screenshots showing: S3 bucket structure, Lambda trigger configuration, the generated report file in S3, and CloudWatch logs